

Benchmark Transition Aware Pricing

Selected topics

London, March 15th 2019

Your speakers today



Dr. Hans Peter Wächter, CFA, FRM

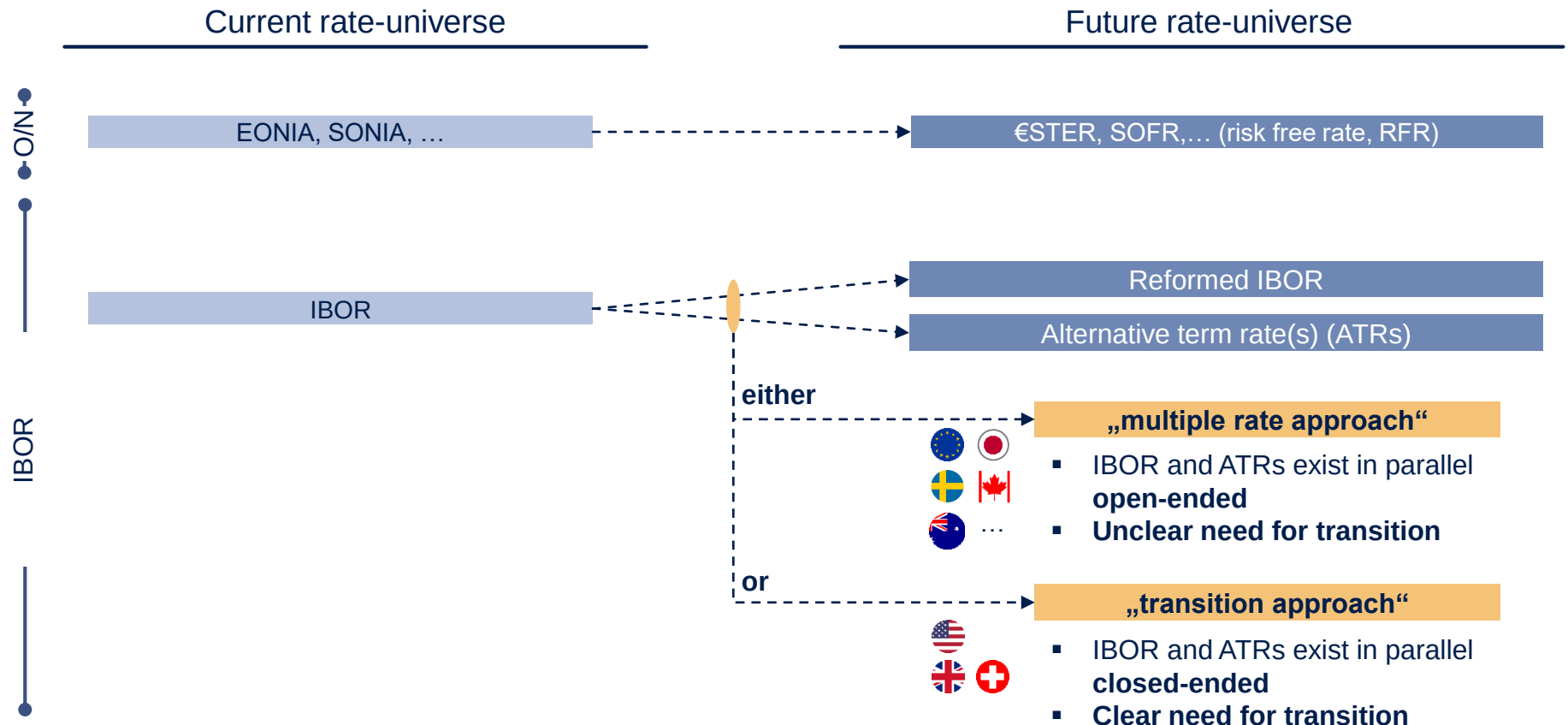
- » Senior Manager at d-fine
- » Focus on topics where valuation intersects with regulatory, accounting and organizational issues, e.g. model governance, IBOR replacement, prudent valuation etc. lately
- » Involvement in various IBOR replacement projects recently



Dr. Holger Plank

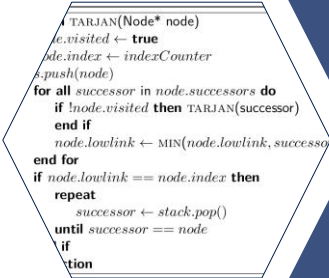
- » Partner at d-fine
- » Focus on xVA and Counterparty Credit Risk including all kinds of regulatory and market induced exposure measures
- » Definition of consistent valuation models for secured and unsecured OTC derivatives on portfolio level as part of many exposure simulation implementation projects
- » Constant work in the quant area since 2003

Currently, the common theme of all changes is the uncertainty of how, when and to where the current rate-universe will be transforming



In all currencies, targets are formulated (and support is given) by legislators and regulators, however solutions are ultimately to be found by the industry itself, **making all transitions „market driven“**.

Loosely following FSB Progress Reports; see also: BIS, *Beyond LIBOR: a primer on the new reference rates*, March 2019.

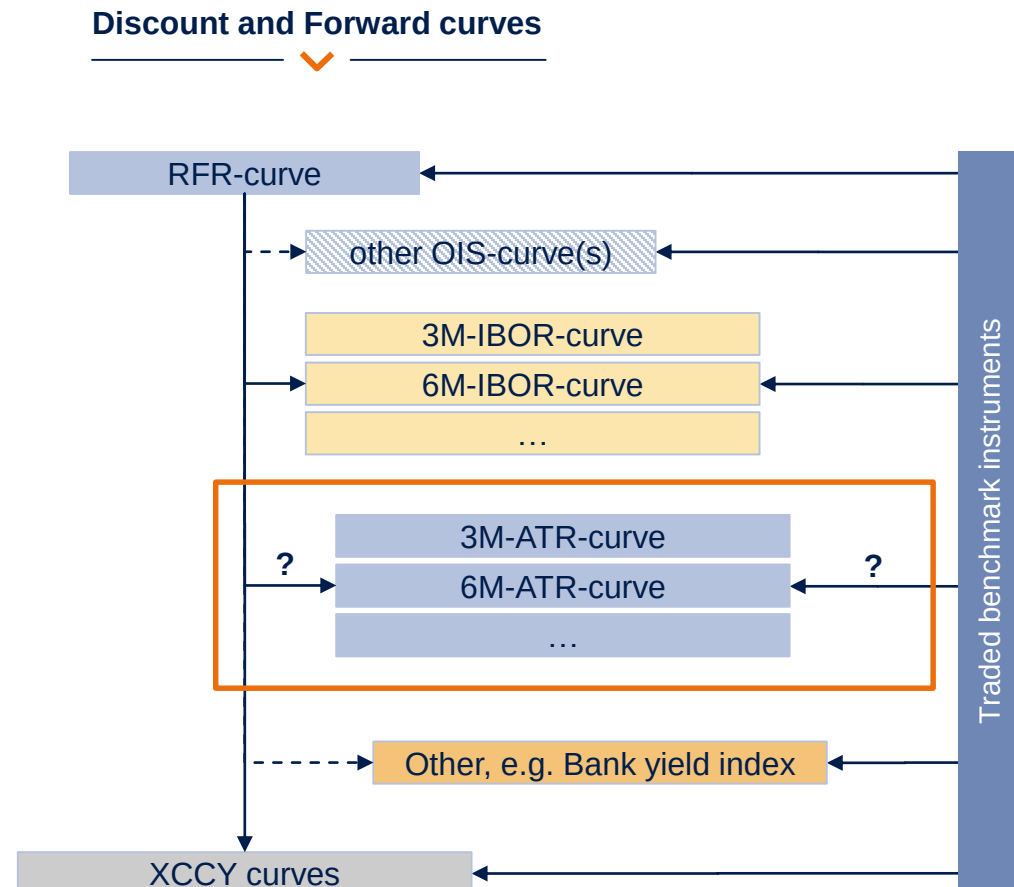
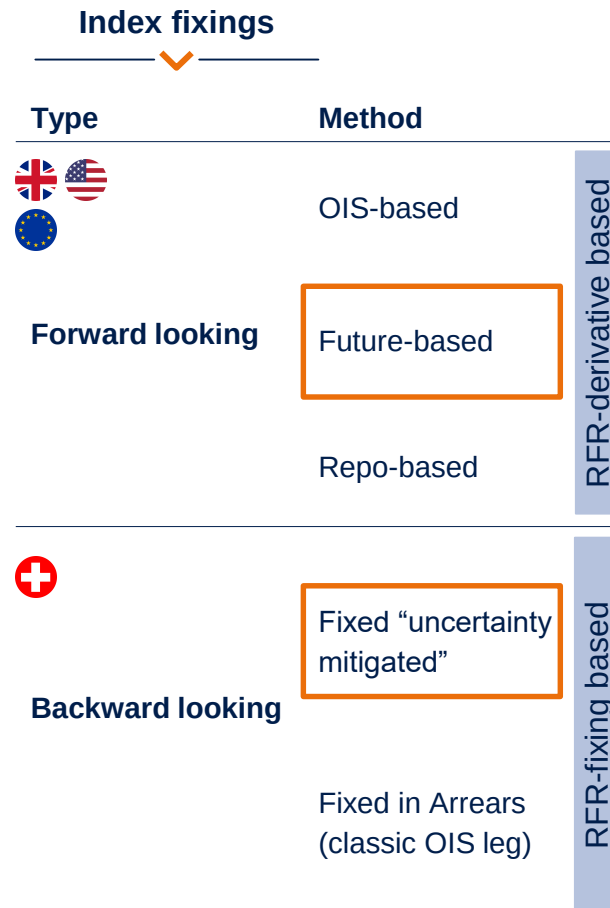


 This talk will cover selected aspects on the topics highlighted: curve building, coupon structures and XVA.

Alternative Term Rates

Selected topics

Alternative term rates (ATR) are expected to be a integral part of the future rate universe – their building principles being currently rather foggy

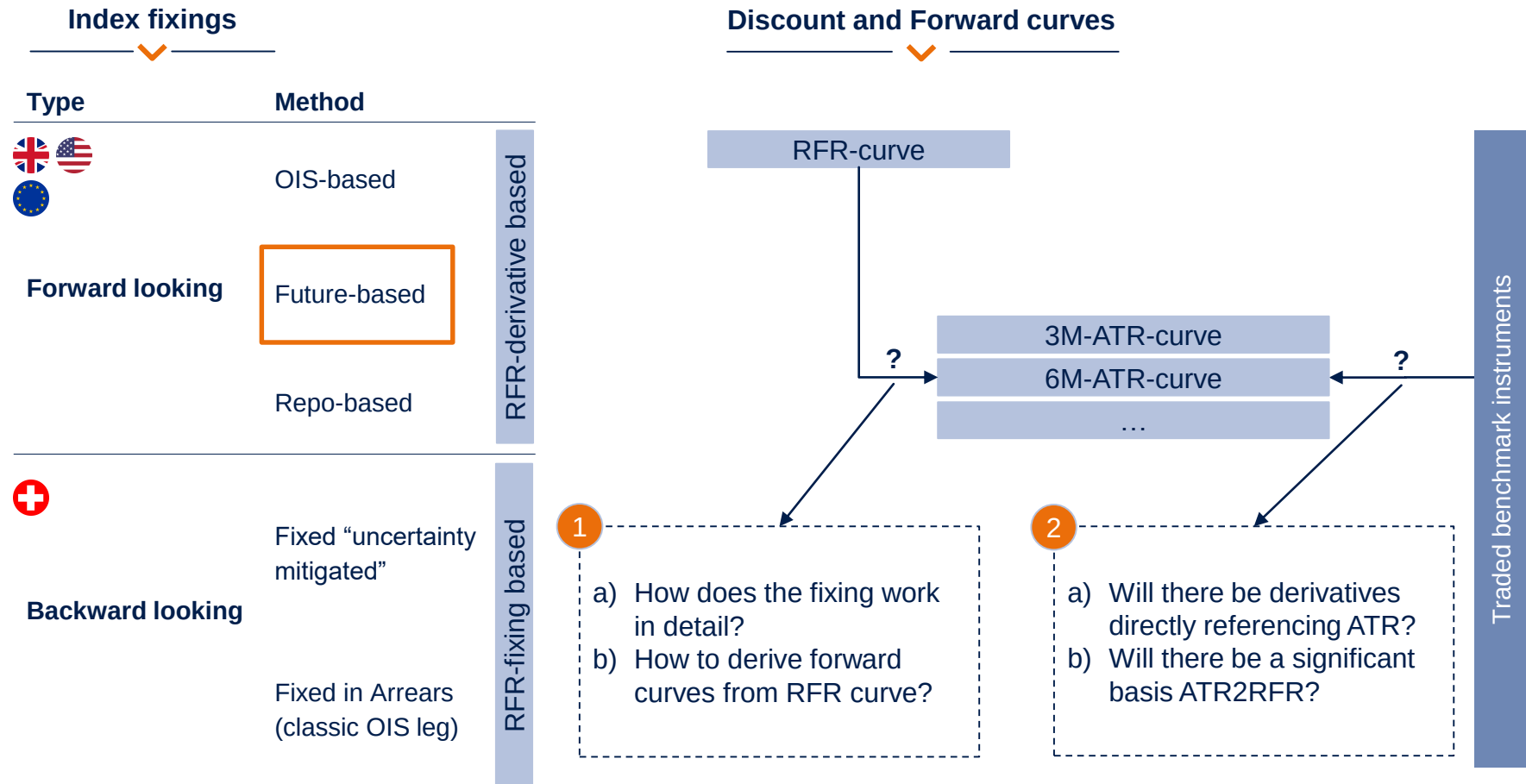


Forward looking and backward looking fixing methods each come with their own complexities; their mitigation will decide on their future feasibility.

Looking forward

Future based methodologies

Future based methodologies commingle the future quotes available over the tenor to a single xM-tenor rate fixed in advance



Question blocks 1) and 2) are (inter alia) decisive for the usability of future based fixings. The questions of the first block will be investigated in the following.

Advanced and detailed accounts on future based methodologies were given by ICE¹ and ACCR² – first of which being discussed in some detail here

Principle

Use future quotes over the tenor ΔT to determine a piece-wise linear implied rate and aggregate this rate to a term rate.

Steps

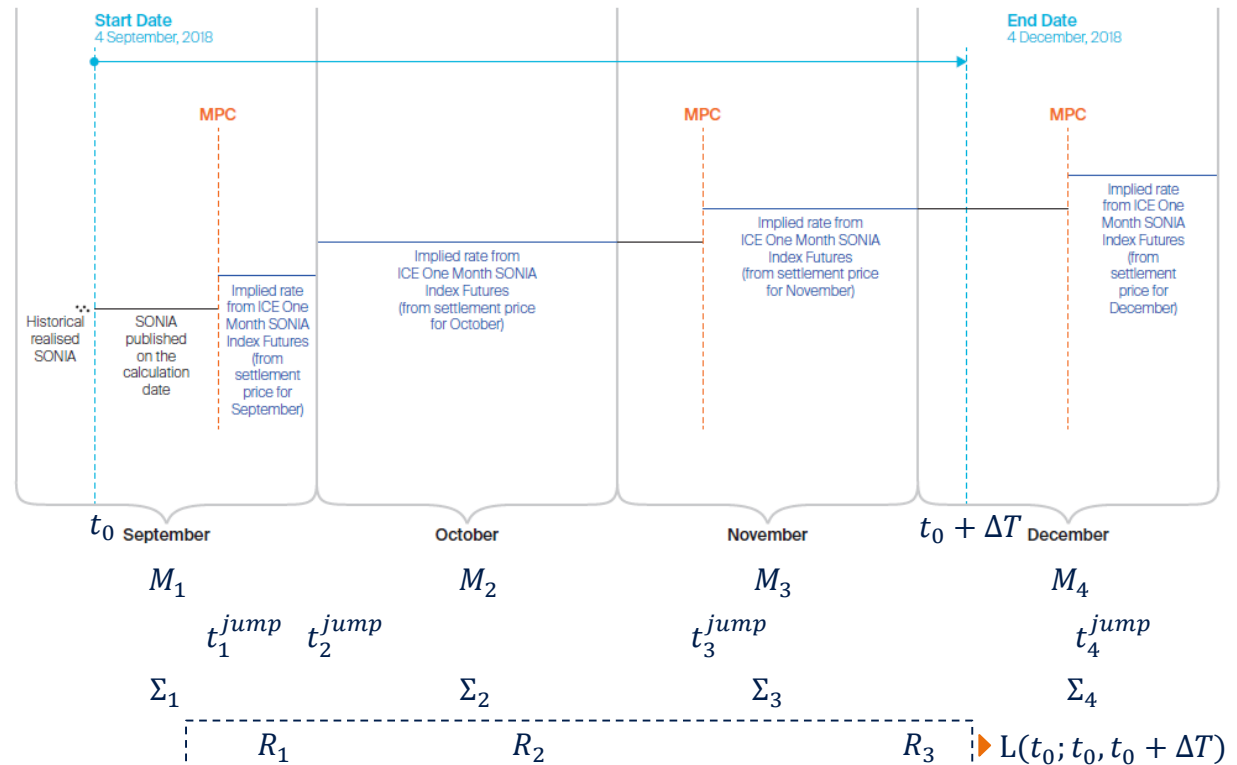
- 1) Determine **relevant months** M_i from start date t_0 to end date $t_0 + \Delta T$
- 2) Determine **jump dates** t_i^{jump} (one per month / MPC date or first day)
- 3) Successively determine **implied rates** R_i by preserving future-quote implied **month sums** Σ_i ; R_i is valid from jump date to jump date
- 4) Compound the implied rates R_i to arrive at the **effective term rate** $L(t_0; t_0, t_0 + \Delta T)$

Month

Jump date

Quote-implied sum

Implied rate



The fixing mechanism needs MPC dates, 1M-future quotes, SONIA fixings in M_1 (and conventions/statics) to produce the effective term rate.

1) Picture taken from ICE BA, *ICE Term Risk Free Rates*, October 2018.
2) ACCR, *Second Report*, March 2018.

The algorithm of ICE can be re-formulated as an inhomogeneous system of linear equations $\underline{\underline{A}} \underline{x} = \underline{b}$; whose solution gives the effective term rate

As there is one jump date per month, the matrix has lower triangular, bi-diagonal form:

$$\begin{array}{c} \text{Months} \rightarrow \\ \leftarrow \text{Impl. Rates} \end{array}
 \begin{pmatrix}
 f_{11} & 0 & 0 & 0 \\
 f_{21} & f_{22} & 0 & 0 \\
 0 & f_{32} & f_{33} & 0 \\
 0 & 0 & f_{43} & f_{44}
 \end{pmatrix}
 \begin{pmatrix}
 R_1 \\
 R_2 \\
 R_3 \\
 R_4
 \end{pmatrix}
 =
 \begin{pmatrix}
 \Sigma'_1 \\
 \Sigma_2 \\
 \Sigma_3 \\
 \Sigma_4
 \end{pmatrix}$$

$\Sigma'_1 = \Sigma_1 - \Sigma^{fix}$: Sum corrected for fixing based sums in first month M_1

Σ_i : Quote-implied whole month sum for month M_i

Matrix elements f_{ij} :

- » Fraction (in calendar days) of month M_i for which rate R_j is applicable
- » Diagonal elements diminish as MPC-date or start date is moved to month end.

Desirable features

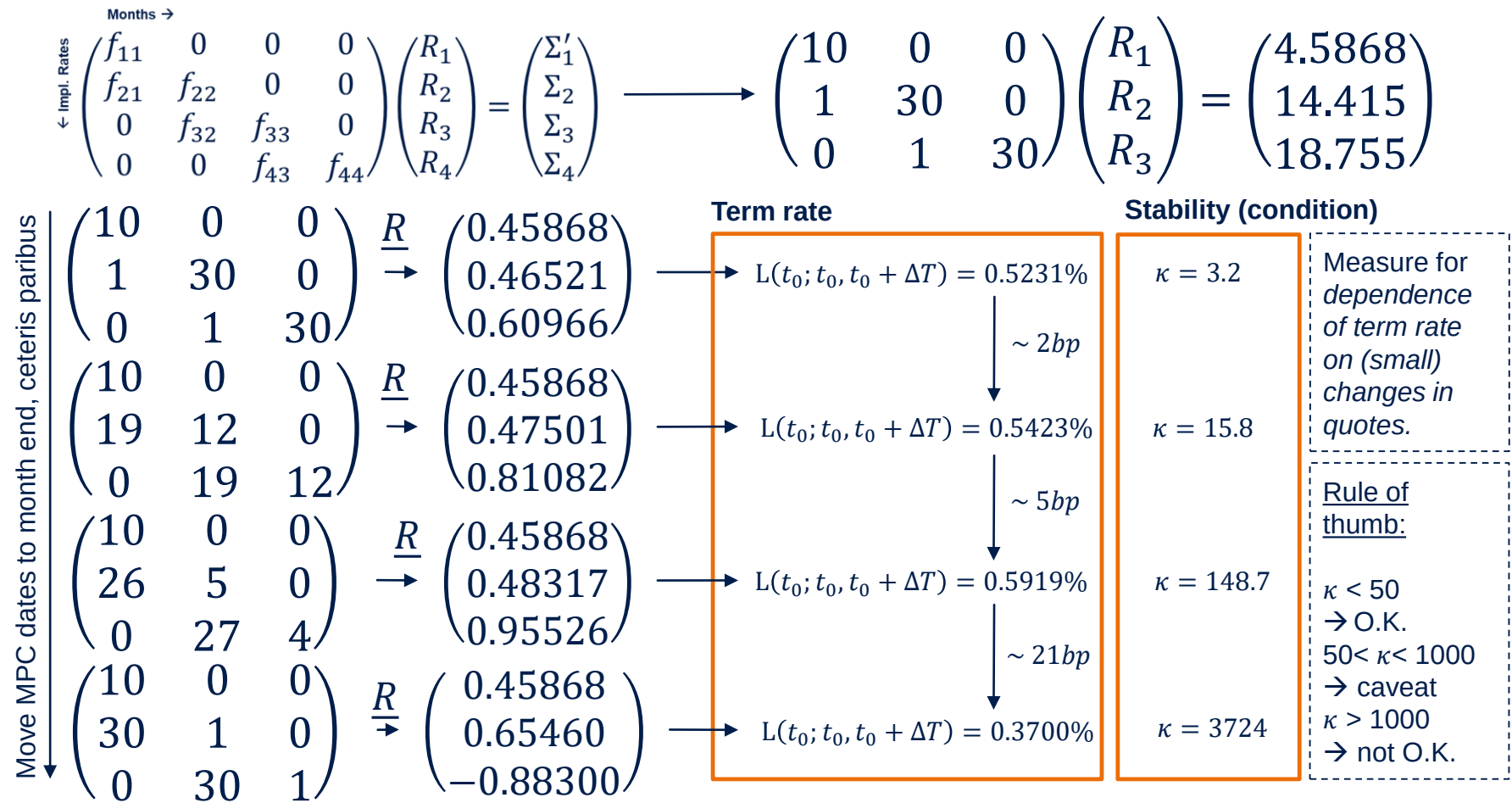
- » **Invertibility:** As $\det(A) = f_{11}f_{22}f_{33}f_{44}$, the matrix is invertible as long as all $f_{ii} \neq 0$. This is warranted by construction.
- » **Stability:** Small deviations in futures prices (or in implied sums) should yield only small deviations in implied rates (and in turn in the effective term rate).
- » **Plausibility:** Implied rates should not jump implausibly.

ARRC proposed an optimization algorithm with penalty function to mitigate these issues.

The re-formulation points out certain caveats, which generically go along with similar matrix based problems: invertibility, stability and plausibility.

Some details omitted for clarity of exposition (e.g. adjustment of number of rows/columns depending on start/end date and jump date of final month). See also ARRC, *Second Report*, March 2018 for some discussion.

The algebraic results can be illustrated by playing with the parameters on example data as of June 8th 2018¹

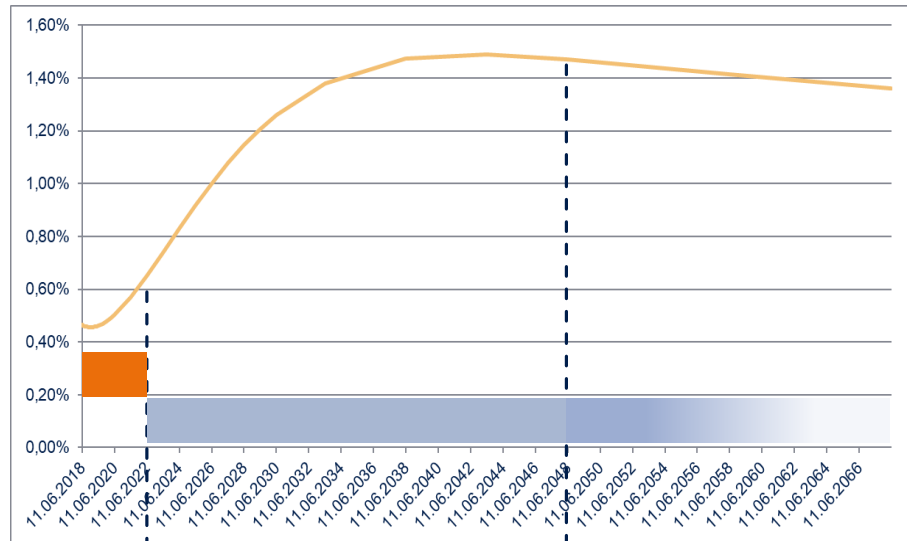


Resulting rates and stability of the problem (measured as condition) significantly depend on the choice of the MPC dates. For calculating forward rates, this yields an unobservable yet important parameter.

1) See ICE BA, *ICE Term Risk Free Rates*, October 2018.
Condition calculated as $\kappa = \|A\| \|A^{-1}\|$; maximum absolute column sum chosen as norm.

An illustrative forward curve can be build using an RFR-curve to project future quotes and overnight fixings – direct forwarding being only a proxy

Illustrative 3M ATR forward curve (no real data used)



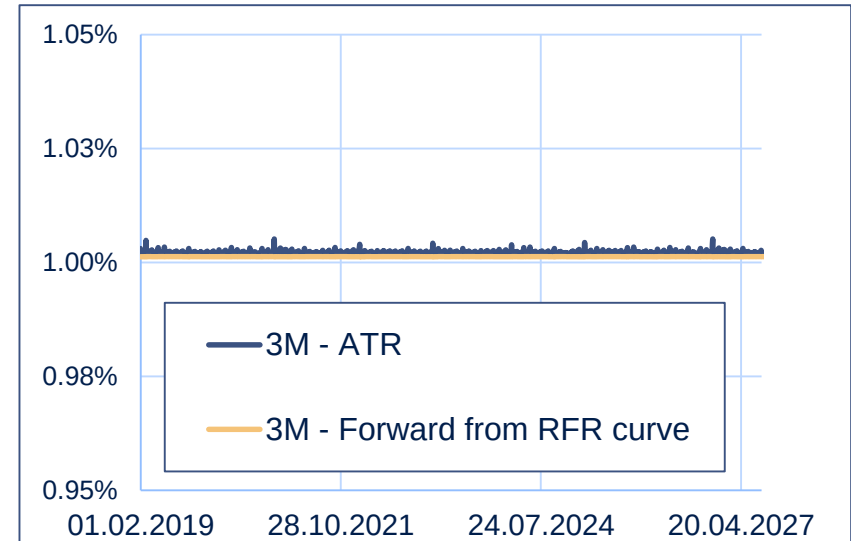
Up to 2Y:

- » Future quotes available
- » MPC dates announced

Up to 30Y:

- » Own ATR derivatives?
- » **Projections from RFR curve needed**

Naïve¹ application of forwarding on flat RFR curve



- » Flat RFR-curve @1,0%; essentially flat 3M-forward curve
 - » ATR-projections done naively¹
 - » MPC-dates chosen close to month end (“horror scenario”)
- **Direct forwarding is only a proxy.**

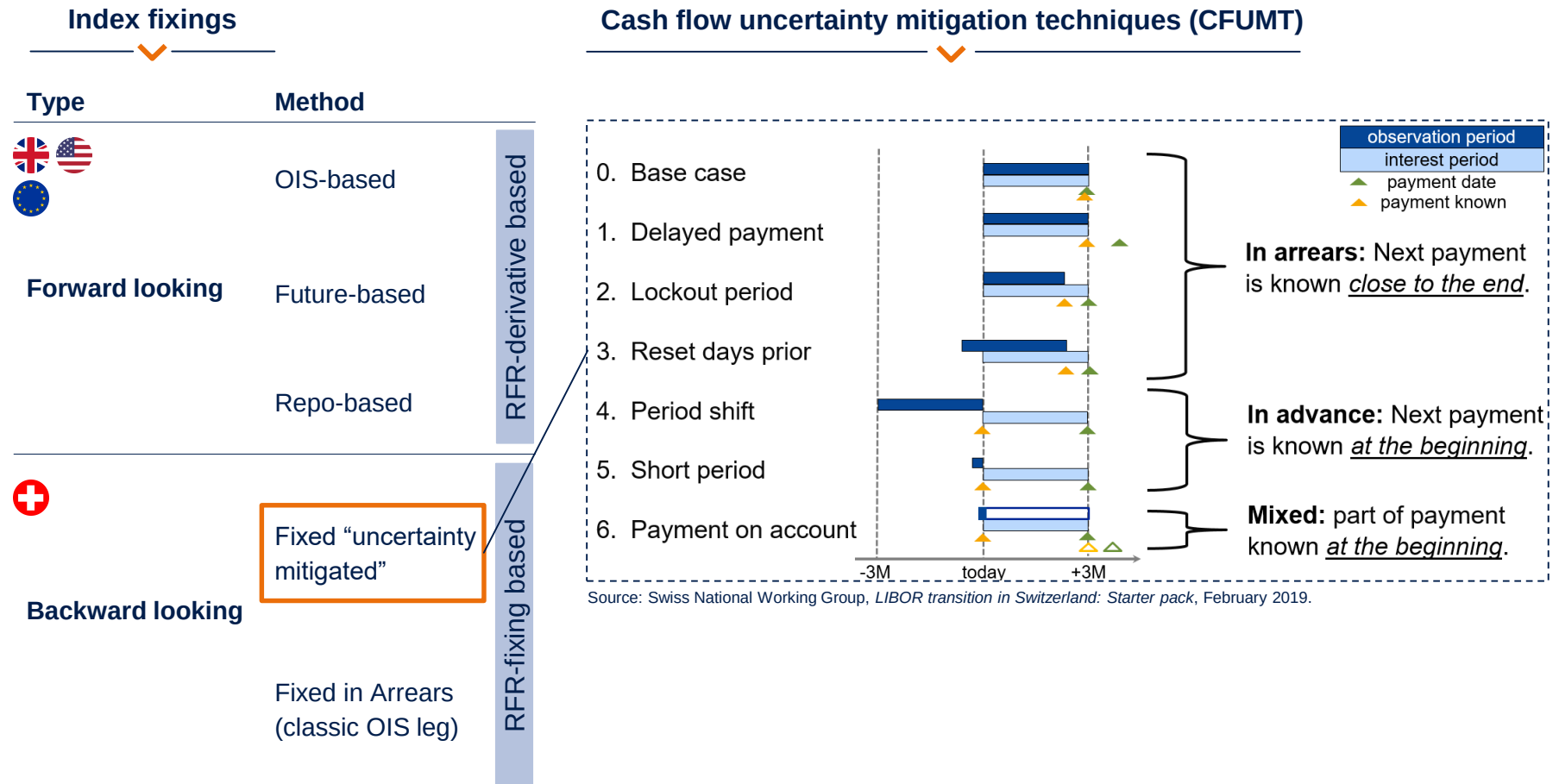
Good news: The mechanism is – in the 3M version shown here – stable apart from edge cases.
Bad news: Mechanism-caused, significant artefacts cannot be ruled out in fixing and forwarding.

1) Assumes $L(t_0; t, t + \Delta T) = \mathbb{E} \left[g \left(\{F_i\}_i; \{Q_j\}_j \right) \right] = g \left(\{\mathbb{E}[F_i]\}_i; \{\mathbb{E}[Q_j]\}_j \right)$ with future overnight fixings F_i and futures quotes Q_j ; g being the algorithm described previously.

Looking backward

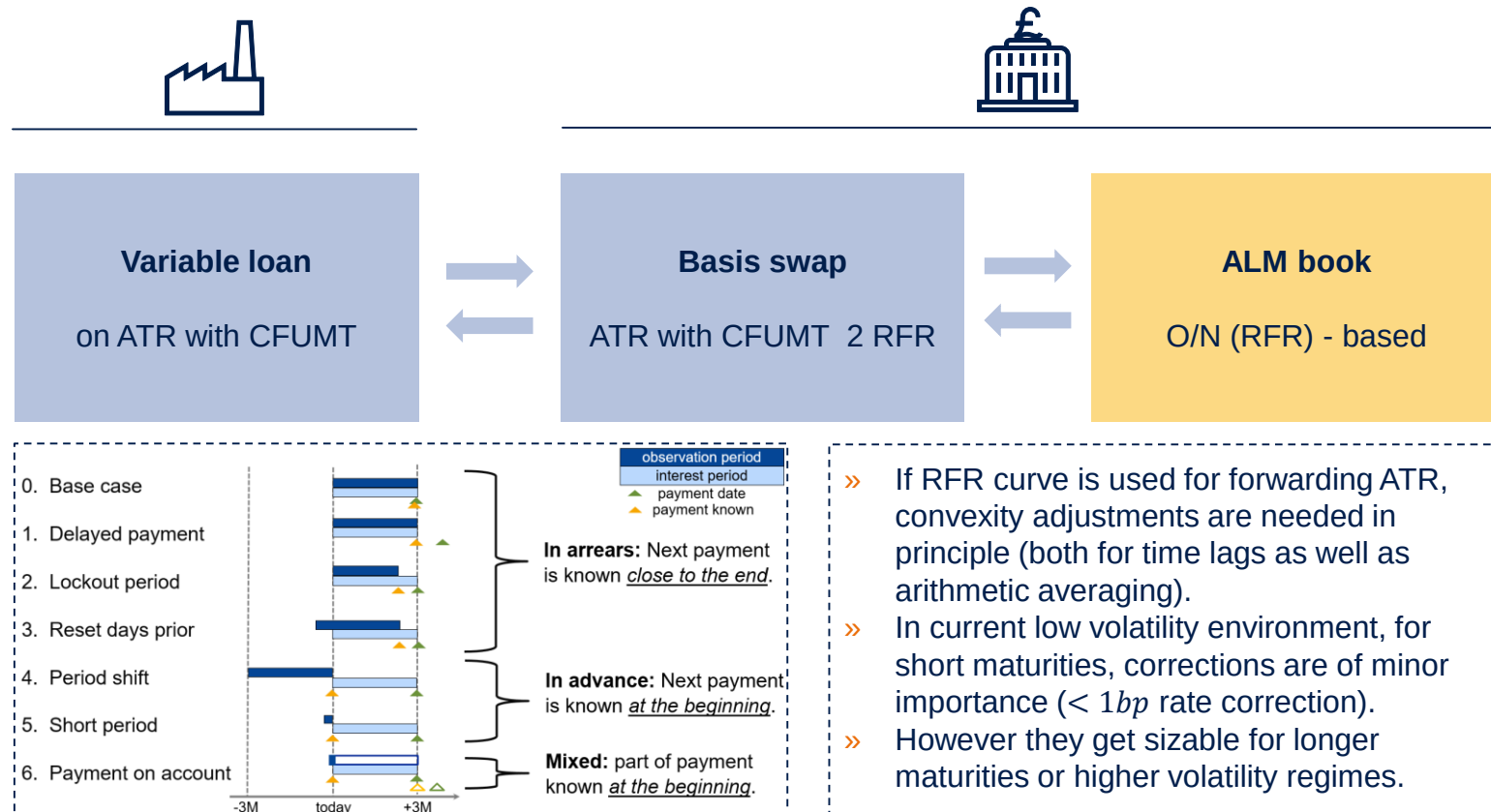
Convexity – the new normal?

The Swiss national working group (NWG) decided to recommend backward looking approaches as ATR



The uncertainty of the period's cashflow inherent to in-arrears fixings can be mitigated using various mitigation techniques (CFUMT).

Each mitigation technique amounts in a non-natural application of the overnight rate (RFR) causing convexity effects (timing adjustments)

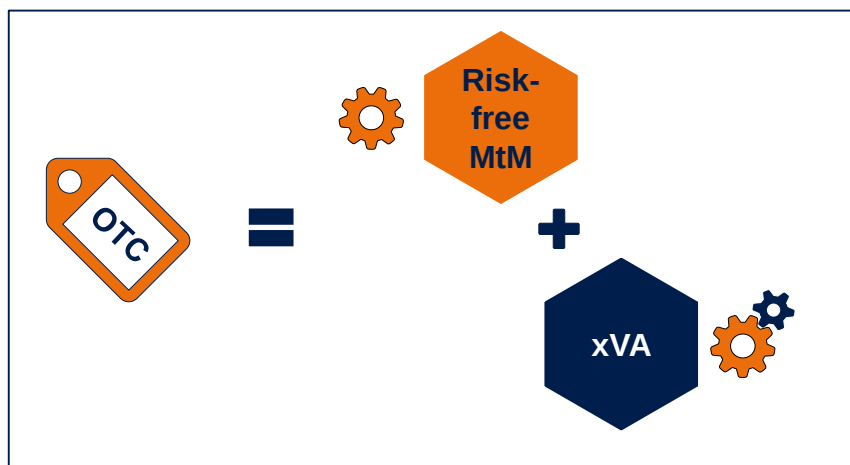


All numbers produced using d-fine's pricing library MoCo.

Convexity effects may well become relevant for ATR with CUMFT in regimes of either high volatility and/or in current environment for long maturities of the loan or swap.

Benchmark Replacements and xVA

Motivation: how are xVA calculation and BMR related to each other?



📌 xVA is a correction to the risk-free price that takes into account the **cost of hedging and running the position** from a bank's point of view.

📌 Each xVA component itself is typically based on expectations of the **distributions of future risk-free prices** of the portfolio the OTC trade belongs to.

📌 **Risk-free pricing** is based on the portfolio's **collateral/PAI curve** (given its collateral currency), typically the according RFR.

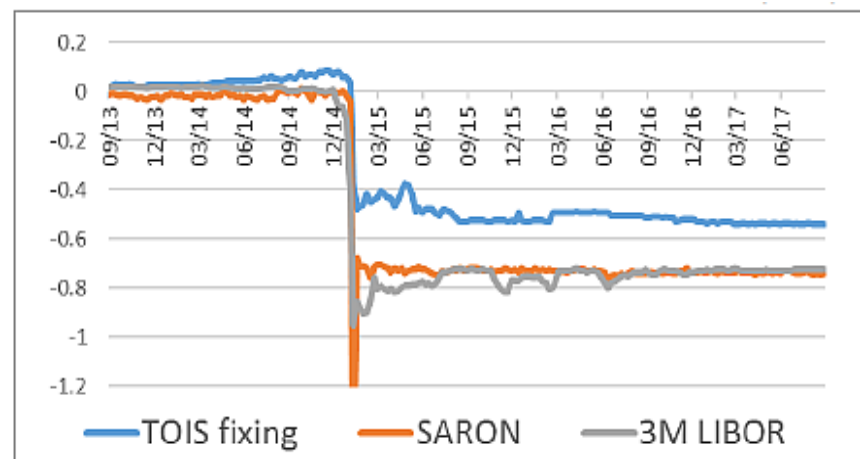
📌 xVA calculation is heavily related to the underlying RFR.

❓ What is going to happen to xVA through the course of the benchmark reforms? Is there a need for current action?

📌 **IFRS 13 / PnL**: each change to an RFR or ATR involved in the portfolio's valuation will lead to a jump in both risk-free MtM and xVA (that are not going to outweigh each other).

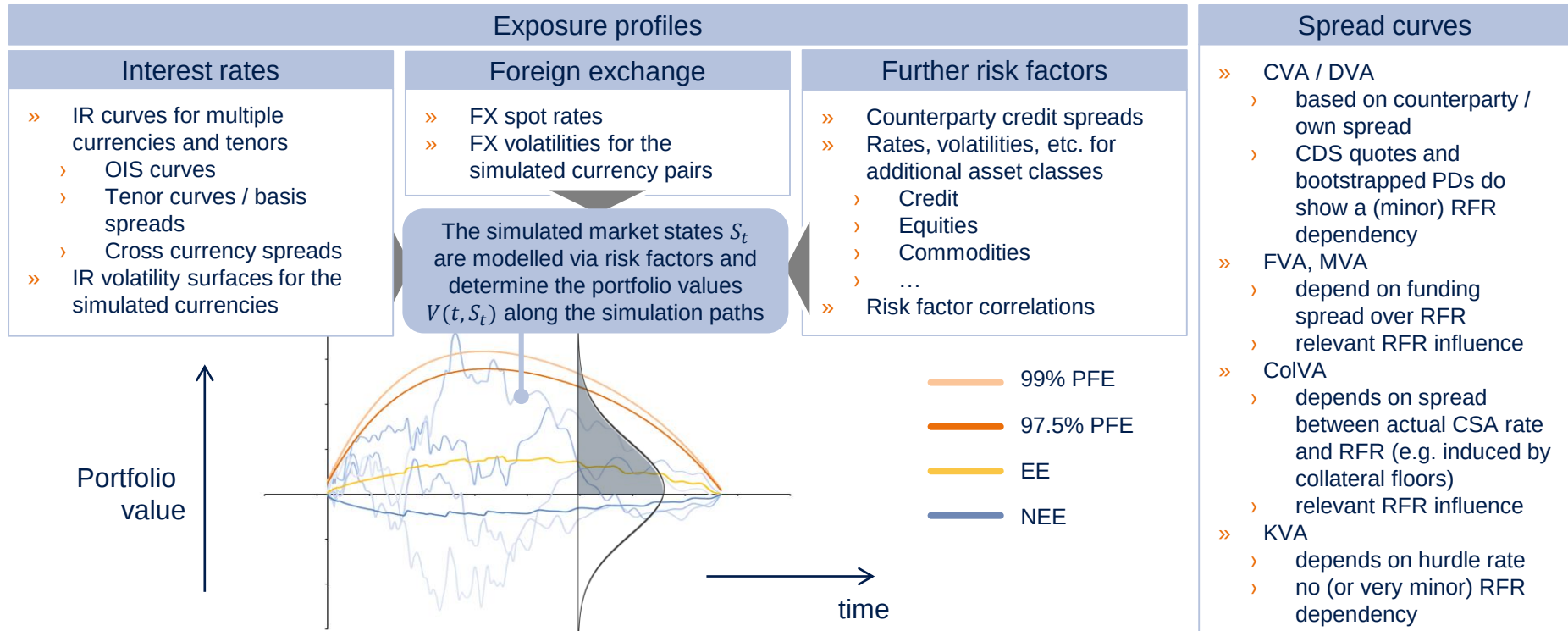
📌 **OTC Pricing**: xVA as a life-time cost of hedging and running the position should include all future costs as foreseen as of today (including counterparty risk, funding, capital, etc.).

📌 Given we have enough information about when and how involved benchmarks are replaced plus according market data, **xVA desks should price in the benchmark reforms into new deals** as the client is *not* going to compensate for it later.



source: Bloomberg
Taken from: M. Bardenhewer, The Adoption of SARON as the Major CHF Risk-Free Benchmark Rate, Zürcher Kantonalbank, 2017

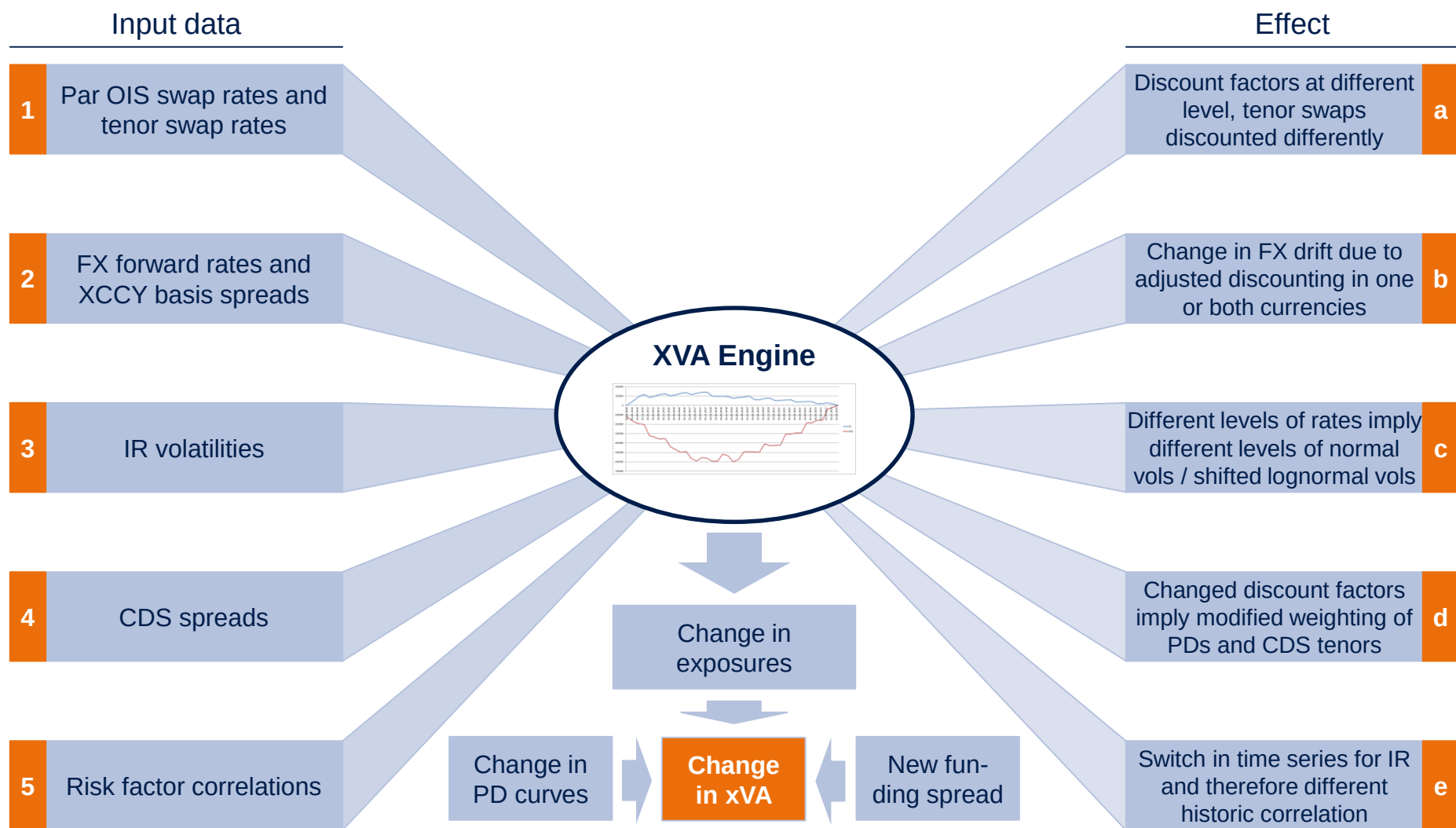
How does an xVA calculation involve RFRs and ATRs?



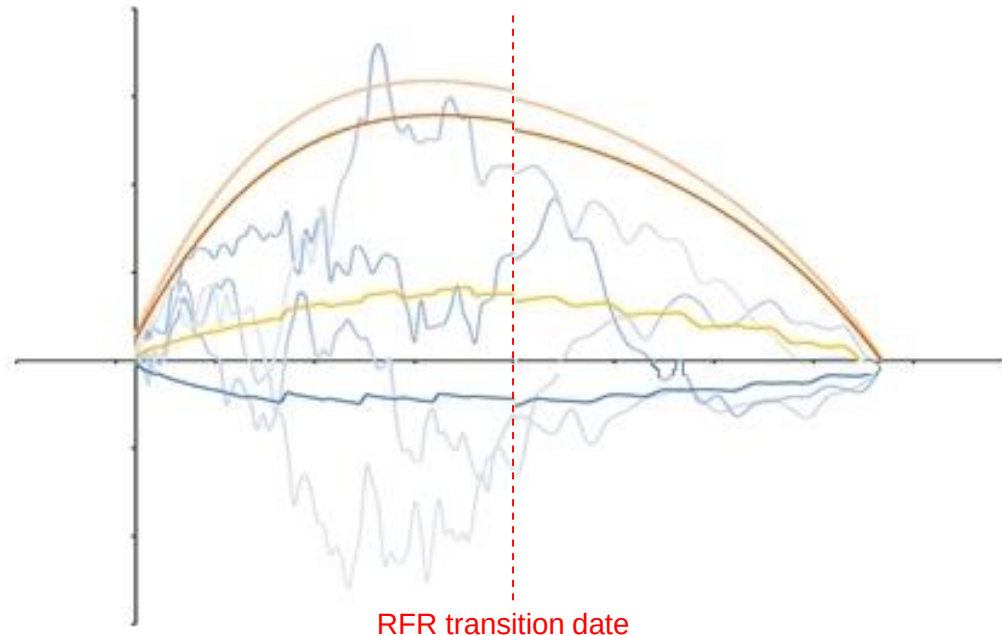
Joint hybrid model on portfolio level

- » All products have to be modelled as part of the same portfolio that is collateralised under a single, perfect CSA.
- » The collateral currency will coincide with the bank's standard reporting currency in most cases.
- » Therefore, XCCY basis spreads get assigned with an important role as part of the exposure simulation: they are needed to adjust discount curves of trades in foreign currency in order to value them as if they are collateralised in the bank's reporting currency.
- » The RFR replacement in the reporting currency will have the biggest impact on the exposure calculation; other RFR replacements as well as new ATRs will have an impact as well, but much more isolated from an impact analysis and validation point of view.

Which inputs to an xVA solution are influenced by the benchmark reform?



Which data is needed as of today to carry out an xVA calculation beyond the RFR/ATR transition date(s)?



✦ A **jump in the RFR** used as CSA / PAI rate for the portfolio's collateral **will trigger a jump in the portfolio value and hence the exposure**.

✦ In order to set up and perform a correct exposure simulation before and after the switch, we **need the curves and vol data before and after the switch** (as of today) plus the expected transition date.

❓ Important questions: which curve to discount with at exposure dates before and after the switch? Does AMC still work?

➡ **Blending (switch between two exposure simulations on the transition date) to be discussed**

- a) there are multiple switch dates
- b) could work as a proxy for blending only from the old to the new CSA rate and ignore other rate transitions
- c) xVA buckets to be aggregated in a post-process

Requirements to the exposure simulation

- » In order to simulate the future MtM distributions correctly, **curves and vols need to be set up to represent the anticipated BMR curve transition** as well as **the future CSA modification** (see next slides).
- » **Market data required to parameterise the simulation may be synthetic** (e.g. consistent with EONIA spot rates until the switch date, €STER forward rates after the switch date) and need to be derived in a pre-processing step.
- » **Required market data may be affected by several transition dates** (e.g. swaption vols by both RFR and ATR switch dates, XCCY spreads by two RFR transitions and the according ATR transitions).
- » **Spread curves used for xVA need to represent the anticipated transition** as well.

Collateralised OTC trading: compensation payment “yes” or “no”?

- » One relevant question with the replacement of TOIS by SARON was whether banks should agree on and **exchange a compensation** payment for the change in the valuation curve. Clearly, that payment would be subject to the renegotiation of the CSA.
- › For a given portfolio with maturity beyond the transition date, consider three different discounting scenarios over the portfolio lifetime:
 - › **Scenario 1: only TOIS discounting**
 - › **Scenario 2: SARON discounting of the trade from** the date $t^{\text{TOIS} \rightarrow \text{SARON}}$ of the collateral switch **onwards** → implies there is a collateral readjustment in Scenario 2 on that day due to the change in value.
 - › From a discounting as well as a collateral earnings perspective, one can see that without compensation payment, **both scenarios are equivalent regarding today's (t_0) value** because the difference in collateral earnings between the transition date and maturity is equal to the amount that needs to be funded to adjust the collateral balance.
 - › **Scenario 3: create a combined curve** that discounts with SARON after the replacement and with a combination of SARON and TOIS for the respective periods before the replacement

$$df_{\text{TOIS} \rightarrow \text{SARON}}(t_0, u) = \exp \left(- \int_{t_0}^{u \wedge t^{\text{TOIS} \rightarrow \text{SARON}}} r_{\text{TOIS}}(s) ds - \int_{t^{\text{TOIS} \rightarrow \text{SARON}}}^{u \vee t^{\text{TOIS} \rightarrow \text{SARON}}} r_{\text{SARON}}(s) ds \right),$$

which obviously leads to a different price at t_0 . So *question* is, if this is correct (as of t_0) or Scenarios 1+2 are correct.

- › *Answer:* check if there is a **CSA** at t_0 **backing Scenario 3**. It is a view of pricing the portfolio taking into account the change in the CSA rate as of a future date (“combined curve”). That actually means a modification to the original CSA which leads or has led to a collateral adjustment (possibly at an earlier day). Even if the CSA covers such a switch, availability of market data is critical for correct pricing.
- › *Equivalently:* if **the parties have agreed** as of t_0 **to pay an compensation** at the day of the rate switch, the compensation amount will equal to the change in value and needs to get priced into Scenario 2 ahead of $t^{\text{TOIS} \rightarrow \text{SARON}}$ by TOIS discounting of the compensation. It is straightforward that in this case there is no MtM jump, hence no collateral readjustment, and **the MtM at t_0 is equal to Scenario 3**.
- » Practical consequences and issues:
 - › You are required to read and **(re-)negotiate** your **CSAs thoroughly and carefully**. Everything is about **timing** of rules, rates, etc.
 - › **Compensation payments may need to be modelled** as part of **collateralised exposure** simulation, subject to the applicable CSA.

Relation of discount rate switch to ColVA

- Let's consider an actual earnings scheme for a collateralised loan (zero bond type) under **Scenario 2** compared to **Scenario 1**:

Scen. 2 (switch known, separate curves)

Day	0.0000	1.0000	2.0000	3.0000	4.0000	5.0000	6.0000	7.0000	8.0000	9.0000	10.0000	
Cashflow											-1000000.0000	Collateral Earnings TOIS + Collateral Readjustment
Discount factor TOIS	1.00014168	1.00012751	1.00011334	1.00009917	1.00008500						1.00000000	
Discount factor SARON						1.00010278	1.00008223	1.00006167	1.00004111	1.00002056	1.00000000	
MtM	-1000141.6767	-1000127.5081	-1000113.3398	-1000099.1716	-1000085.0036	-1000102.7831	-1000082.2256	-1000061.6686	-1000041.1120	-1000020.5558	-1000000.0000	
Close-Out Premium	1000141.6767	1000127.5081	1000113.3398	1000099.1716	1000085.0036	1000102.7831	1000082.2256	1000061.6686	1000041.1120	1000020.5558	1000000.0000	Plain Sums (no discounting)
Collateral Start of Day	0.0000	-1000127.5081	-1000113.3398	-1000099.1716	-1000085.0036	-1000070.8358	-1000082.2256	-1000061.6686	-1000041.1120	-1000020.5558	-1000000.0000	Collateral Adjustment(day 5) using TOIS
Collateral End of Day	-1000141.6767	-1000127.5081	-1000113.3398	-1000099.1716	-1000085.0036	-1000102.7831	-1000082.2256	-1000061.6686	-1000041.1120	-1000020.5558	0.0000	-31.9495
Collateral Readjustment	-1000141.6767	0.0000	0.0000	0.0000	0.0000	-31.9472	0.0000	0.0000	0.0000	0.0000	1000000.0000	-173.6239
Interest on Collateral	0.0000	-14.1686	-14.1684	-14.1682	-14.1680	-14.1678	-20.5575	-20.5570	-20.5566	-20.5562	-20.5558	-173.6239
MtM Gain/Loss		14.1686	14.1684	14.1682	14.1680	-17.7794	20.5575	20.5570	20.5566	20.5562	20.5558	141.6767
												-31.9472
Difference scenarios 1 vs. 2						-31.9472	6.3899	6.3897	6.3894	6.3892	6.3890	0.0000

- Our intuition on xVA theory may tell us the last line (from day 6 onwards) looks like a **ColVA** valued at time $t' = t^{\text{TOIS} \rightarrow \text{SARON}}$.
- And in fact, referring to *Piterbarg, Funding beyond discounting* (Risk February 2010), we recall equality of equations (3) and (5) there – normally used to translate the difference between LIBOR and OIS discounting into an FVA type integral for unsecured OTC –

$$\begin{aligned}
 V(t) &= E \left[\exp \left(- \int_t^T r_F(u) du \right) V(T) + \int_t^T \exp \left(- \int_t^u r_F(v) dv \right) (r_F(u) - r_C(u)) C(u) du \right] \\
 &= E \left[\exp \left(- \int_t^T r_C(u) du \right) V(T) - \int_t^T \exp \left(- \int_t^u r_C(v) dv \right) (r_F(u) - r_C(u)) (V(u) - C(u)) du \right].
 \end{aligned}$$

- We apply this formula to the case $t = t'$, $r_C = r_{\text{SARON}}$ and $r_F = r_{\text{TOIS}}$ in the case of $V = V_{\text{SARON}} = C = C_{\text{SARON}}$ and obtain

$$V_{\text{TOIS}}(t') - V_{\text{SARON}}(t') = \text{ColVA}_{\text{TOIS} \rightarrow \text{SARON}}(t^{\text{TOIS} \rightarrow \text{SARON}}) \equiv \int_{t'}^T (r_{\text{SARON}}(s) - r_{\text{TOIS}}(s)) E[\text{df}_{\text{TOIS}}(t', s) C_{\text{SARON}}(s)] ds.$$

Obviously, this formula holds for all $t' \leq u \leq T$ instead of t' and can be extended to $u \leq t'$ if the SARON short rate is set equal to the TOIS short rate there*.

- Practical applications of this formula need to be discussed. Obviously, the ColVA implies using Monte Carlo simulation, volatility information (often with substantial simplifications like absence of smiles) and may therefore include more sources of uncertainty.

* Note that this results requires the volatility of our xVA model to be invariant under the change of benchmark curves and the two curves should have a deterministic spread.

Uncollateralised (=client) case under the RFR transition

- » *Question:* how does the case of a switch in the RFR (standard collateral rate) translate to the **uncollateralised client OTC business**?
 - » First observation: there will be (most likely) **no compensation** between the parties for a change in discounting as there is no CSA.
 - » Second observation: if we already know about the change in the collateral rate as of the RFR transition date t_{ccy}^{OIS} , we have to **incorporate any impact on the risk-free price and xVA into the pre-deal price!**
 - » This is because we cannot renegotiate the contract without a formal reason: the change in the float leg index at t_{ccy}^{IBOR} (ATR transition) will cause a change in the contract, but a change in the discount curve does not since it is typically not determined in the contract.

- » *Solution:* **apply well known principles from xVA theory!**

- » xVA theory tells us that for a uncollateralised derivative and a default-free client we have (apply Piterbarg to the case $C = 0$)

$$V(t) = E \left[\exp \left(- \int_t^T r_F(u) du \right) V(T) \right] = E \left[\exp \left(- \int_t^T r_C(u) du \right) V(T) - \int_t^T \exp \left(- \int_t^u r_C(v) dv \right) (r_F(u) - r_C(u)) V(u) du \right]^*,$$

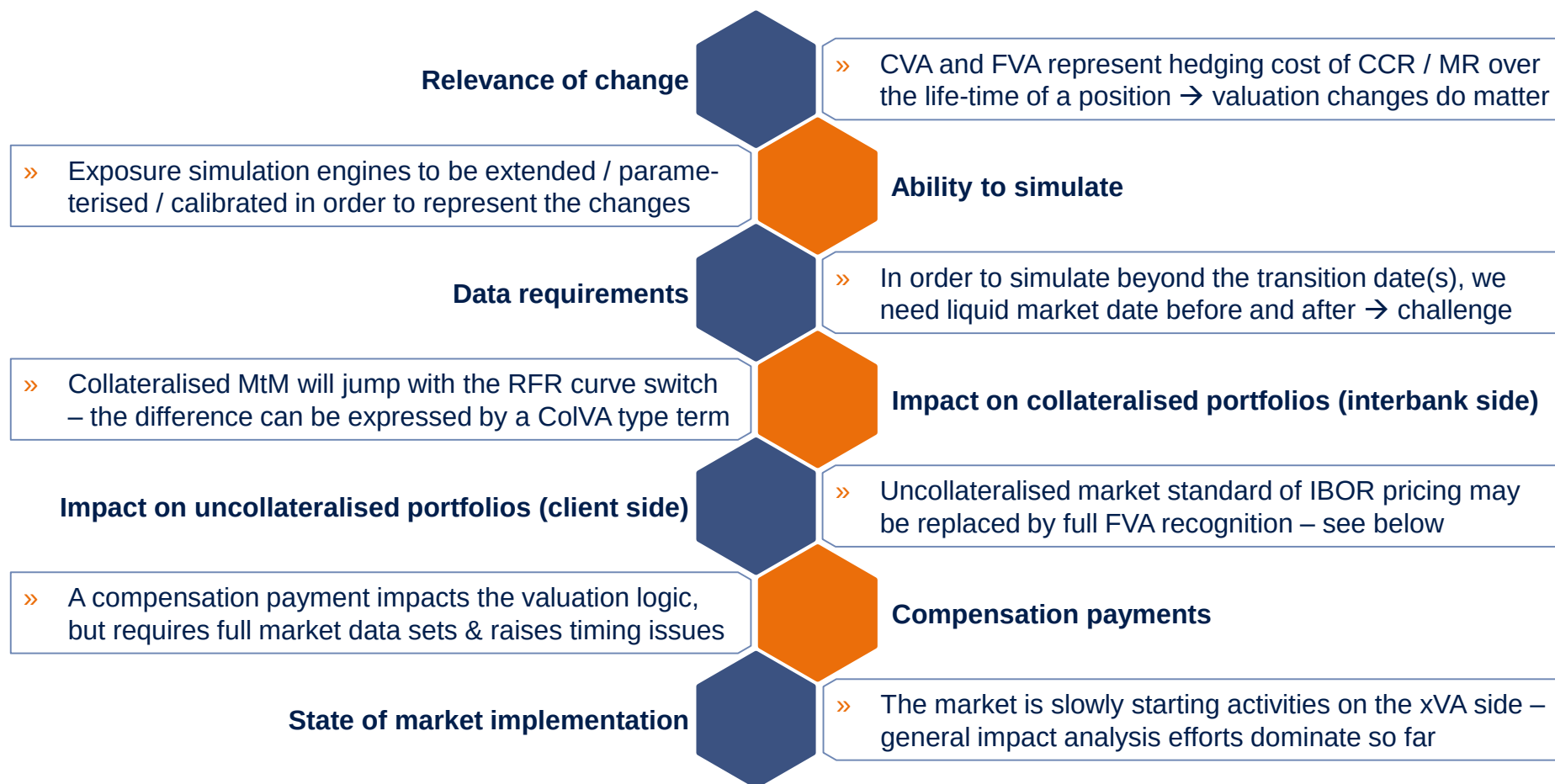
or, equivalently, with straightforward definitions of the risk-free value w.r.t. the funding and collateral curves

$$V^F(t) = V^C(t) - FVA(t).$$

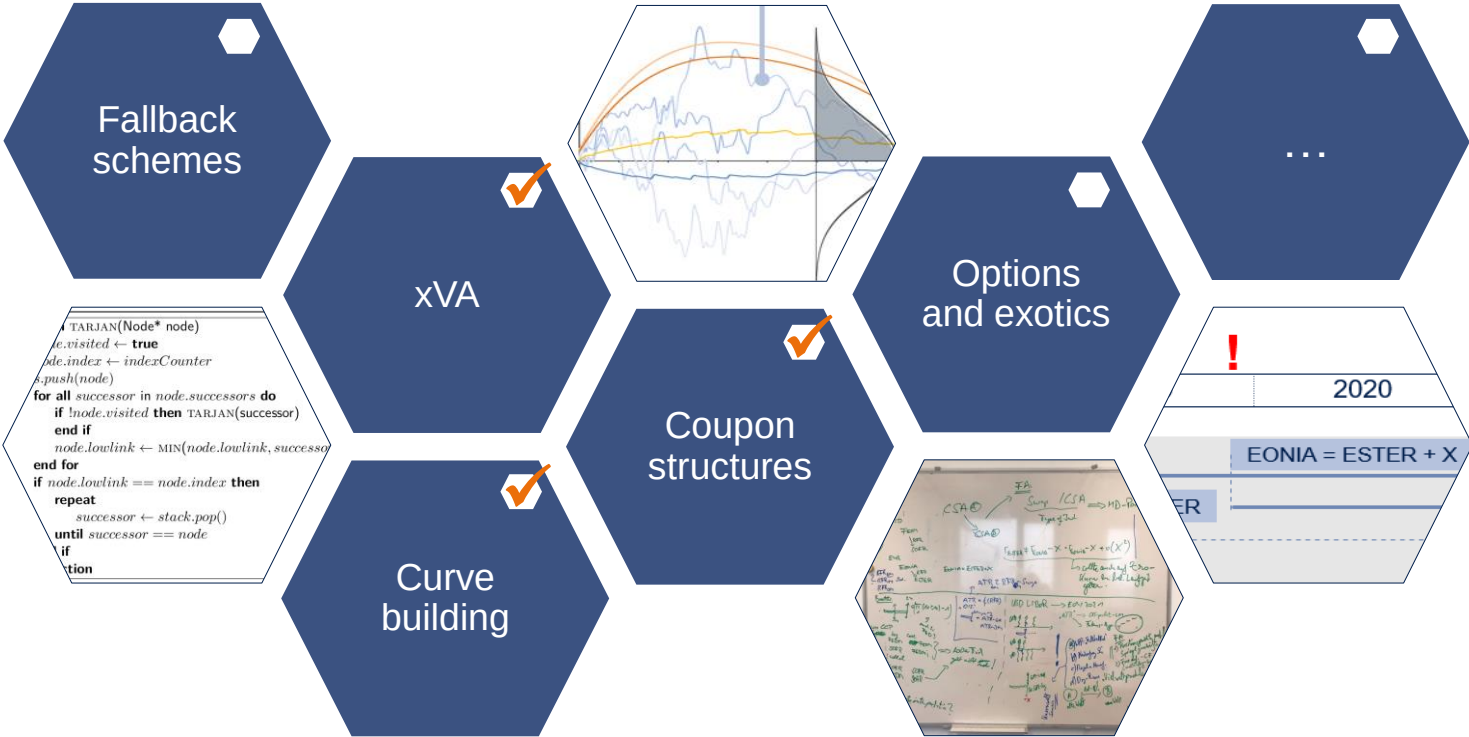
- » Since we know how to value $V^C(t)$ from the previous discussion (Scenarios 1 to 3), we therefore **find the uncollateralised price by calculating the funding cost and benefit of the position over its lifetime** including the known switch in the collateral rate.
- » As an immediate consequence, we are faced with two additional questions:
 - » How to determine the bank's funding curve before and after the switch of the benchmark rate?
 - » What about the market standard of calculating the uncollateralised price via IBOR discounting?
- » Final answers are to be determined by the market.
 - » If we relate the bank's **funding cost** to the **bond market**, this is probably unlikely to jump at the date of the collateral rate switch (same for the float rate transition?) → to be decided: is the **funding curve** going to change or the **funding spread** w.r.t. the RFR?
 - » As soon as we have decided which assumptions to put on the funding spread: in case we still want to stick to the method of tenor curve discounting of uncollateralised trades, we will have to define (i.e. bootstrap) an **interim, combined IBOR-alike tenor curve** that is compliant with the formula above.

* A discrete compensation payment will add an additional term to the r.h.s. (i.e. FVA) that must not be ignored!

Conclusion: main topics concerning benchmark reforms and xVA



Proper representation of the benchmark reforms in the xVA system is key for OTC pricing and accounting. Availability of data and potential implementation topics in the engine require a transition phase.



 This talk covered selected aspects on the topics highlighted: Many more discussions to come!

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